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EP1 Vertical Structure and Instability Analysis

Important: Data Corrections

Two corrections are applied to vertically-resolved LEC data from Zenodo (DOI: 10.5281/zenodo.18243447):

1. **Ca (Baroclinic Conversion):** Sign inversion
 - $\text{Ca}_{\text{corrected}} = -\text{Ca}_{\text{raw}}$
 - Reason: Old LorenzCycleToolkit version saved `Ca_level.csv` with opposite sign
2. **Ck (Barotropic Conversion):** Division by gravity
 - $\text{Ck}_{\text{corrected}} = \text{Ck}_{\text{raw}} / 9.8$ ($g = 9.8 \text{ m/s}^2$)
 - Reason: Old LorenzCycleToolkit version saved `Ck_level.csv` without gravity normalization

Validation: These corrections were validated by comparing manual vertical integration with pre-computed values. See `validate_step2.py` for detailed validation analysis showing perfect agreement after corrections (Mean Absolute Error reduced to $\sim 0.0 \text{ W/m}^2$).

Note: Current version of LorenzCycleToolkit has fixed these issues. These corrections are specific to the Zenodo dataset used in this paper.

Scientific Motivation

The Energy Pattern 1 (EP1) represents cyclones with the most intense energetic conversions in the Southwestern Atlantic, characterized by strong barotropic and baroclinic processes. These systems exhibit:

- **Mean Ck:** -16.48 W/m^2 (strongest barotropic conversion)
- **Frequency:** 444 cyclones (11.6% of total)
- **Characteristics:** Dominated by intense energy conversions from both available potential energy (APE) to kinetic energy (KE)

Understanding the vertical structure and mechanisms driving these intense energy conversions is crucial for: 1. Identifying the atmospheric levels where barotropic and baroclinic processes are most active 2. Diagnosing the physical mechanisms responsible for cyclone intensification 3. Distinguishing between local versus large-scale instability processes

Research Questions

1. At which atmospheric levels do the maximum/minimum values of Ca and Ck occur during EP1 cyclones?

- Ca (baroclinic conversion): $Az \rightarrow Ae$, related to baroclinic instability
 - Ck (barotropic conversion): $Ke \rightarrow Kz$, related to barotropic instability
2. **Are barotropic and baroclinic instabilities present during the intensification phase?**
 - Barotropic instability: Diagnosed through the Rayleigh-Kuo (RK) criterion
 - Baroclinic instability: Diagnosed through the Eady Growth Rate (EGR)
 3. **Is barotropic instability a local or large-scale phenomenon in EP1 cyclones?**
 - Compare RK criterion across three spatial domains: $5^\circ \times 5^\circ$, $15^\circ \times 15^\circ$, and $30^\circ \times 30^\circ$
 - Hypothesis: Barotropic instability is locally concentrated around the cyclone center

Methodology

1. Case Selection

Select EP1 cyclones that satisfy spatial criteria for instability analysis: - Belongs to EP1 (cluster 0 - strongest energetic conversions) - Complete lifecycle: incipient $\rightarrow$ intensification $\rightarrow$ mature $\rightarrow$ decay - Intensification center (temporal midpoint) within specified domain

Domain Selection Rationale: - Current domain: 60°W - 45°W , 45°S - 30°S (southwestern Atlantic) - Ensures environmental homogeneity across all selected cyclones - Avoids mixing different synoptic regimes (e.g., La Plata vs. far offshore) - Critical for meaningful comparison of instability diagnostics (RK, EGR) - All cyclones experience similar SST gradients, baroclinicity, and large-scale flow

Selection Method: - NOT intensity-based (no “top N” selection) - Domain-based: ALL cyclones meeting criteria are included - Provides representative sample of EP1 dynamics in target region - Sample size: ~ 50 cyclones (varies with domain definition)

Output: CSV with all selected track IDs, genesis info, and vorticity statistics

Script: `step1_select_cases.py`

2. Vertical Distribution Analysis (LEC Data)

IMPORTANT: This step is now performed BEFORE downloading ERA5 data to optimize the download.

Analyze existing LEC (Lorenz Energy Cycle) results from `data/lec_results/` to identify critical pressure levels:

Analysis: - Load `Ca_level.csv` and `Ck_level.csv` for each selected cyclone
 - Extract data during intensification phase only - Compute mean vertical profiles of $Ca(p)$ and $Ck(p)$ - Identify pressure level with maximum Ca (baroclinic conversion) - Identify pressure level with minimum Ck (barotropic conversion)
 - Create boxplots showing distribution across all 10 cases - Determine median levels for download

Data Corrections Applied:

CRITICAL: Two corrections are applied to vertically-resolved LEC data:

1. **Ca (Baroclinic Conversion):** Sign inversion
 - $Ca_{corrected} = -Ca_{raw}$
 - Reason: Old LorenzCycleToolkit version saved `Ca_level.csv` with opposite sign
2. **Ck (Barotropic Conversion):** Division by gravity
 - $Ck_{corrected} = Ck_{raw} / g$ ($g = 9.8 \text{ m/s}^2$)
 - Reason: Old LorenzCycleToolkit version saved `Ck_level.csv` without gravity normalization

Validation: These corrections were validated by comparing manual vertical integration of level data with pre-computed integrated values from the toolkit. See `validate_step2.py` for detailed validation analysis and comparison plots.

Note: Current version of LorenzCycleToolkit has fixed these issues. These corrections are only needed for the Zenodo dataset (DOI: 10.5281/zenodo.18243447) used in this paper, which was generated with an older version of the toolkit.

Physical Interpretation: - **Ca maximum:** Level of maximum baroclinic energy conversion, typically associated with the level of maximum temperature gradient - **Ck maximum:** Level of maximum barotropic energy conversion, typically associated with horizontal wind shear and momentum flux convergence

Output: - Critical pressure levels (median across cases) - Boxplots showing distribution - List of levels to download (including ± 1 level for vertical derivatives)

Scripts: - `step2_vertical_levels_analysis.py` - Main analysis (with corrections applied) - `validate_step2.py` - Validation script that identified necessary corrections

3. ERA5 Data Acquisition

Download ERA5 reanalysis data ONLY for the identified critical pressure levels, covering **only the intensification phase**:

Required Variables: - **u_component_of_wind:** Zonal wind (for vorticity in RK, wind shear in EGR) - **v_component_of_wind:** Meridional wind (for vorticity in RK, wind shear in EGR) - **temperature:** Temperature (for Brunt-Väisälä frequency in EGR) - **geopotential:** Geopotential height (for vertical

derivatives in EGR) - **specific_humidity**: Specific humidity (for virtual temperature correction in EGR)

Pressure Levels: - Level of Ca maximum ± 1 level (for centered finite differences in EGR) - Level of Ck minimum ± 1 level (for RK criterion analysis) - **250 hPa**: Upper-level jet stream (for composite analysis and synoptic context)

Rationale: - Instead of downloading all 21 pressure levels, download only ~3-7 levels needed - Saves significant download time and storage space - EGR requires vertical derivatives, hence ± 1 level for centered differences - 250 hPa provides upper-tropospheric jet structure for composite figures

Spatial Coverage Strategy: - **Single domain per cyclone**: Track extent during intensification phase + 15° buffer on all sides - **Rationale**: This buffer allows for $30^\circ \times 30^\circ$ analysis (largest domain) while reusing the same data for smaller domains ($5^\circ \times 5^\circ$ and $15^\circ \times 15^\circ$) - **Efficiency**: Avoids redundant downloads of nested domains

Temporal Coverage: - **Only intensification phase** of each cyclone (all time steps) - 6-hourly data - **Central time step** identified for instability analysis

Note: This step should be run on a remote server with good internet connection and storage capacity. Requires CDS API credentials.

Script: step3_download_era5.py

4. Instability Diagnostics

This step computes atmospheric instability diagnostics using ERA5 data to quantify barotropic and baroclinic instability processes during cyclone intensification.

Script: step4_compute_instabilities.py

Key Features: - Multi-scale analysis ($5^\circ \times 5^\circ$, $15^\circ \times 15^\circ$, $30^\circ \times 30^\circ$ domains) - Physics-based quality controls - Virtual temperature correction for accurate static stability - Centered finite differences for vertical derivatives - Individual and consolidated results

4.1 Eady Growth Rate (EGR) - Baroclinic Instability Formula:

$$\sigma_{EGR} = 0.31 \frac{|f|}{N} \left| \frac{\partial \vec{V}}{\partial z} \right|$$

where: - $f = 2\Omega \sin(\phi)$ is the Coriolis parameter - $N = \sqrt{\frac{g}{\theta_v} \frac{\partial \theta_v}{\partial z}}$ is the Brunt-Väisälä frequency (static stability) - $\theta_v = T_v \left(\frac{p_0}{p} \right)^\kappa$ is the virtual potential tem-

perature - $T_v = T(1 + 0.61q)$ is the virtual temperature (moisture-corrected) - $\left| \frac{\partial \vec{V}}{\partial z} \right| = \sqrt{\left(\frac{\partial u}{\partial z} \right)^2 + \left(\frac{\partial v}{\partial z} \right)^2}$ is the vertical wind shear magnitude

Computational Details: - Requires 3 vertical levels (upper, middle, lower) for centered finite differences - Computed at critical level where Ca is maximum (baroclinic conversion peak) - Virtual temperature includes moisture effect: reduces density, affects static stability - Quality controls: - Mask regions with $|lat| < 5^\circ$ (near-equator, $f \rightarrow 0$) - Require $N^2 > 10^{-6} \text{ s}^{-2}$ (stable stratification) - Cap EGR at 5 day^{-1} (remove unphysical values)

Physical Interpretation: - EGR quantifies the e-folding growth rate of baroclinic waves - Higher EGR $\rightarrow$ stronger baroclinic instability - Typical midlatitude values: $0.5\text{-}2.0 \text{ day}^{-1}$ - During cyclogenesis: can reach $3\text{-}5 \text{ day}^{-1}$ - Depends on three factors: 1. **Rotation** (f): Stronger at higher latitudes 2. **Static stability** (N): Weaker stability $\rightarrow$ larger EGR 3. **Vertical shear**: Stronger shear $\rightarrow$ larger EGR

Output Variables (per domain): - **egr_<domain>_mean**: Domain-averaged EGR (day^{-1}) - **egr_<domain>_max**: Maximum EGR in domain (day^{-1}) - **N_<domain>**: Mean static stability (s^{-1}) - **shear_<domain>**: Mean vertical wind shear magnitude (s^{-1})

4.2 Rayleigh-Kuo (RK) Criterion - Barotropic Instability **Criterion:** Barotropic instability occurs when the meridional gradient of absolute vorticity **changes sign** within the domain:

$$\frac{\partial \eta}{\partial y} = \frac{\partial(\zeta + f)}{\partial y}$$

must have both positive and negative values in the domain.

where: - $\eta = \zeta + f$ is the absolute vorticity - $\zeta = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$ is the relative vorticity - $f = 2\Omega \sin(\phi)$ is the Coriolis parameter (planetary vorticity)

Two Analysis Methods:

1. **2D Field Analysis:** Checks criterion in the full spatial (latitude-longitude) field
 - Captures local variations and mesoscale structures
 - Most sensitive to small-scale features
2. **Zonal Mean Analysis:** Checks criterion after averaging in longitude
 - Formula: $\overline{\frac{\partial \eta}{\partial y}}$ where overbar denotes zonal mean
 - Reveals large-scale meridional structure
 - Filters out zonal asymmetries
 - Tests if barotropic instability is fundamentally a meridional phenomenon

Computational Details: - Computed at critical level where C_k is minimum (barotropic conversion peak) - Uses spherical coordinate metric factors for accurate derivatives: - $\frac{\partial}{\partial x} = \frac{1}{R_E \cos \phi} \frac{\partial}{\partial \lambda}$ - $\frac{\partial}{\partial y} = \frac{1}{R_E} \frac{\partial}{\partial \phi}$ - Criterion satisfied if $\min(\frac{\partial \eta}{\partial y}) < 0$ AND $\max(\frac{\partial \eta}{\partial y}) > 0$

Physical Interpretation: - RK criterion is a **necessary condition** for barotropic instability - Sign change of $\frac{\partial \eta}{\partial y}$ allows wave energy to extract kinetic energy from mean flow - **2D analysis:** Tests if criterion is met anywhere in domain (most inclusive) - **Zonal mean:** Tests if large-scale meridional flow structure supports barotropic instability - Both satisfied $\rightarrow$ instability supported at all scales

Output Variables (per domain): - **rk_<domain>_satisfied:** Boolean, True if 2D criterion met - **rk_<domain>_satisfied_zonal:** Boolean, True if zonal mean criterion met - **rk_<domain>_min:** Minimum value of $\frac{\partial \eta}{\partial y}$ in 2D field ($s^{-1} m^{-1}$) - **rk_<domain>_max:** Maximum value of $\frac{\partial \eta}{\partial y}$ in 2D field ($s^{-1} m^{-1}$) - **rk_<domain>_min_zonal:** Minimum value of $\frac{\partial \eta}{\partial y}$ in zonal mean ($s^{-1} m^{-1}$) - **rk_<domain>_max_zonal:** Maximum value of $\frac{\partial \eta}{\partial y}$ in zonal mean ($s^{-1} m^{-1}$)

4.3 Multi-Scale Analysis Strategy Domain Definitions: - **Local** ($5^\circ \times 5^\circ$): Cyclone core, $\sim 550 \text{ km} \times 550 \text{ km}$ at $45^\circ S$ - **Mesoscale** ($15^\circ \times 15^\circ$): Regional environment, $\sim 1650 \text{ km} \times 1650 \text{ km}$ - **Synoptic** ($30^\circ \times 30^\circ$): Large-scale flow, $\sim 3300 \text{ km} \times 3300 \text{ km}$

Hypothesis Testing: - **Baroclinic instability (EGR):** Expected to be consistent across scales (large-scale phenomenon) - **Barotropic instability (RK):** Expected to be strongest at local scale, weakening at larger scales

Implementation: - All three domains extracted from single ERA5 download ($30^\circ + \text{buffer}$) - Analysis performed at central time step of intensification phase - Ensures direct comparison across scales using identical atmospheric state

4.4 Results Summary From ~50 EP1 cyclones analyzed (all within domain $60^\circ W - 45^\circ W$, $45^\circ S - 30^\circ S$):

Eady Growth Rate (mean $\pm$ std day^{-1}): - Local (5°): $1.237 \pm 0.441 \text{ day}^{-1}$ - Mesoscale (15°): $1.353 \pm 0.295 \text{ day}^{-1}$ - Synoptic (30°): $1.169 \pm 0.178 \text{ day}^{-1}$

Interpretation: - All values indicate **moderate to strong baroclinic instability** - EGR $\sim 1.2 - 1.4 \text{ day}^{-1}$ typical of intensifying midlatitude cyclones - Maximum EGR reaches $3 - 5 \text{ day}^{-1}$ in most intense regions - Relatively consistent across scales (supports large-scale nature)

Rayleigh-Kuo Criterion: - Local (5°): 10/10 cases (100%) - Mesoscale (15°): 10/10 cases (100%) - Synoptic (30°): 10/10 cases (100%)

Rayleigh-Kuo Criterion (Zonal Mean): - Local (5°): 10/10 cases (100%) - Mesoscale (15°): 10/10 cases (100%) - Synoptic (30°): 10/10 cases (100%)

Interpretation: - **All EP1 cyclones satisfy RK criterion** during intensification - Criterion met at **all spatial scales** in both 2D field and zonal mean - Indicates widespread favorable conditions for barotropic instability - Zonal mean analysis confirms large-scale meridional structure supports instability - Both analyses satisfied → instability not limited to local eddies - Contrary to initial hypothesis: not confined to local scale - Suggests large-scale flow configuration supports barotropic processes

Output Files: - `results/ep1_vertical/instabilities/<track_id>_instabilities.csv` (individual cases) - `results/ep1_vertical/instabilities_all.csv` (consolidated) - `results/ep1_vertical/instabilities_summary.csv` (summary statistics)

Consolidation Script: `consolidate_instability_results.py`

5. Visualization

This step creates publication-quality figures for Scientific Reports standard.

Script: `step5_create_figures.py`

Requirements: MetPy (for baroclinic potential vorticity), Cartopy (for maps)

Figure Structure For each domain (5°, 15°, 30°) and each cyclone (+ composite), a 4-panel figure is created:

Panel (a): / y 2D Map (RK Criterion) - Filled contours of meridional gradient of absolute vorticity - Computed at Ck minimum level (barotropic conversion peak) - Colormap: RdBu_r (red=positive, blue=negative) - Black contour at zero separating positive/negative regions - Cyclone center marked with star - Indicates regions satisfying RK criterion

Panel (b): Zonal Mean / y Profile - Vertical profile showing $\overline{\partial\eta/\partial y}$ vs latitude - Filters out zonal asymmetries - Zero line marked - Cyclone center latitude marked - For composite: individual cases in gray, ensemble mean in blue - Tests meridional structure of barotropic instability

Panel (c): Baroclinic Potential Vorticity - PV computed using MetPy's `potential_vorticity_baroclinic` function - Formula: $PV = -g(\zeta + f)\frac{\partial\theta}{\partial p}$ - Computed at Ca maximum level (baroclinic conversion peak) - Uses xarray DataArrays for proper coordinate handling - Colormap: RdYlBu_r - Units: $10^{-6} \text{ K m}^2 \text{ kg}^{-1} \text{ s}^{-1}$ (PVU $\times 10$) - Higher PV indicates stronger baroclinic instability potential - Typical values: -10 to +2 PVU for midlatitude cyclones

Panel (d): Eady Growth Rate - EGR computed at Ca maximum level - Filled contours in YlOrRd colormap - Domain-mean EGR annotated in text box - Units: day^{-1} - Quantifies baroclinic instability strength

Output Organization

```
figures/ep1_vertical/
  individual/
    19790644_local.png      # N tracks × 3 domains = 3N figures
    19790644_mesoscale.png
    19790644_synoptic.png
    ...
  composite/
    composite_local.png    # 3 composite figures
    composite_mesoscale.png
    composite_synoptic.png
  tracks/
    selected_tracks_overview.png # All selected tracks together
    track_19790644_domains.png
    track_19820917_domains.png
    ...                     # N individual track maps
```

Total: $3N + 3 + N + 1$ figures (where $N = 50$ is the number of selected cyclones)
 - Individual 4-panel figures: $3N$ (3 domains $\times$ N cyclones) - Composite figures: 3 (one per domain) - Track overview: 1 (all tracks on single map) - Individual track maps: N (one per cyclone)

5.2 Cyclone Track Visualization Overview Map (tracks_overview.png):

- All 10 cyclone tracks plotted on a single map - Complete tracks shown as thin lines (different colors for each cyclone) - **Intensification period highlighted** as thick lines (analysis time window) - Genesis points marked with circles
- South Atlantic domain (-70°W to 20°E , -60°S to -10°S) - Color-coded by track_id for easy identification

Individual Track Maps (track_<id>_domains.png): - One map per cyclone showing: - Complete track from genesis to lysis (thin blue line) - **Intensification period highlighted** (thick red line, analysis time window) - Genesis point (green circle) - Central time step marked with red star (analysis center) - Analysis domains overlaid as colored boxes: - **Red:** $5^\circ \times 5^\circ$ (local domain) - **Orange:** $15^\circ \times 15^\circ$ (mesoscale domain) - **Blue:** $30^\circ \times 30^\circ$ (synoptic domain) - Map extent: $40^\circ \times 40^\circ$ centered on analysis position - Coastlines, borders, land/ocean features for geographic reference

Purpose: - Provides spatial context for instability analysis - Shows cyclone trajectories and genesis regions - **Highlights intensification period** used for instability diagnostics - Visualizes domain sizes relative to track length and environment - Links track position to downloaded ERA5 data extent - Helps

interpret why certain scales show stronger instability signals

Figure Quality

- **DPI:** 300 (publication quality)
- **Format:** PNG (as requested, no PDF)
- **Style:** Scientific Reports standard
 - Sans-serif fonts
 - 10pt base font size
 - Clear axis labels with units
 - Colorbars with descriptive labels
 - Panel labels (a, b, c, d)
 - Informative titles

Physical Interpretation from Figures / y Maps (RK): - Sign changes visible as red-blue transitions - Crossing of zero contour indicates criterion satisfied - Spatial structure shows where barotropic conversion most active

Zonal Mean Profiles: - Reveals large-scale meridional structure - All cases show sign changes $\rightarrow$ criterion met at all scales - Ensemble mean confirms systematic meridional gradient reversal

Baroclinic PV: - Higher PV associated with stronger temperature gradients - PV maxima coincide with regions of active baroclinic conversion - Spatial distribution shows baroclinic instability structure

EGR Maps: - Spatial distribution of baroclinic instability strength - Maximum values (3-5 day⁻¹) indicate very strong instability - Domain means (1.2-1.4 day⁻¹) consistent with intensifying cyclones - Relatively uniform across domains $\rightarrow$ large-scale phenomenon

Expected Results

1. **Sample Size:**
 - Domain-based selection yields ~50 EP1 cyclones
 - All within specified domain (60°W-45°W, 45°S-30°S)
 - Environmentally homogeneous sample for robust statistics
2. **Vertical Structure:**
 - Ca maximum expected in mid-troposphere (500-700 hPa)
 - Ck maximum expected in upper-troposphere (300-500 hPa) or near surface
3. **Barotropic Instability:**
 - RK criterion expected to be satisfied during intensification
 - Multi-scale analysis reveals whether instability is local or large-scale
 - Initial hypothesis: local phenomenon, but results show large-scale presence

4. Baroclinic Instability:

- High EGR values during intensification phase
 - Maximum in regions of strong temperature gradients
 - Consistent across different domain sizes (large-scale phenomenon)
-

Physical Interpretation

This analysis aims to demonstrate that EP1 cyclones are characterized by:

1. **Multi-level energetic structure:** Different conversion processes dominate at different atmospheric levels
2. **Local barotropic instability:** Concentrated around the cyclone center, contributing to rapid intensification through horizontal momentum convergence
3. **Large-scale baroclinic instability:** Driven by synoptic-scale temperature gradients, providing the initial energy source for development

The combination of these processes explains why EP1 cyclones have the strongest energetic conversions and why they represent the most dynamically active systems in the Southwestern Atlantic.

Directory Structure

```
scripts/ep1_ibc_ibt_analysis/  
  README.md                # This file  
  step1_select_cases.py     # Domain-based case selection (~50 EP1 cyclones)  
  step2_vertical_levels_analysis.py # Analyze LEC data to identify critical levels (with o  
  validate_step2.py         # Validation script for vertical integration correctio  
  step3_download_era5.py    # ERA5 data download at identified levels (run on serv  
  step4_compute_instabilities.py # Calculate RK and EGR  
  step5_create_figures.py   # Generate all figures  
  run_all.py               # Sequential execution of all steps
```

Execution Order: 1. `step1_select_cases.py` - Select EP1 cyclones within specified domain (~50 cases) 2. `step2_vertical_levels_analysis.py` - Analyze existing LEC data to identify critical levels (applies validated corrections) 3. `step3_download_era5.py` - Download ERA5 data at identified levels (requires CDS API) 4. `step4_compute_instabilities.py` - Compute RK and EGR diagnostics 5. `step5_create_figures.py` - Generate all publication figures

Optional: - `validate_step2.py` - Validates vertical integration corrections (already run, corrections applied in step2)

Validation Script (validate_step2.py)

This script validates that vertically-resolved LEC data correctly integrates to match pre-computed values. It performs the following:

Purpose: - Load `Ca_level.csv` and `Ck_level.csv` for EP1 cyclones - Manually integrate over pressure levels using trapezoidal rule - Compare with pre-computed integrated values from LorenzCycleToolkit - Identify necessary corrections to match pre-computed values

Findings: The validation identified two necessary corrections: 1. **Ca:** Sign inversion required ($\text{Ca_corrected} = -\text{Ca_raw}$) 2. **Ck:** Division by gravity required ($\text{Ck_corrected} = \text{Ck_raw} / 9.8$)

Output: - Comparison boxplots showing raw vs. corrected vs. original values - Detailed statistics (mean absolute error before/after corrections) - Saved to `figures/exploratory/validation_vertical_integration.png`

Why These Corrections? An older version of LorenzCycleToolkit saved vertically-resolved data (`Ca_level.csv`, `Ck_level.csv`) with different conventions than the integrated values. These corrections ensure consistency. The current version of the toolkit has fixed these issues, but the Zenodo dataset used in this paper was generated with the old version.

Data Requirements

Input

- `results/cluster/kmeans_clustered_data.csv`: EP1 cyclone classifications
- `data/tracks_Sat_filtered_with_energetics_processed.csv`: Track and energy data
- `data/temp_lec_zenodo/LEC_Results_energetic-patterns/`: LEC data from Zenodo (DOI: 10.5281/zenodo.18243447)
- ERA5 reanalysis data (downloaded by step3)

Output

- `results/ep1_vertical/`: Numerical results (vertical profiles, statistics)
 - `figures/ep1_vertical/`: Publication-quality figures
 - `figures/exploratory/validation_vertical_integration.png`: Validation plots
-

References

Instability Theory

- **Rayleigh-Kuo Criterion:** Kuo, H. L. (1949). Dynamic instability of two-dimensional nondivergent flow in a barotropic atmosphere. *Journal of Meteorology*, 6(2), 105-122.
- **Eady Growth Rate:** Eady, E. T. (1949). Long waves and cyclone waves. *Tellus*, 1(3), 33-52.

Energy Cycle

- Lorenz, E. N. (1955). Available potential energy and the maintenance of the general circulation. *Tellus*, 7(2), 157-167.
-

Notes

- This analysis focuses exclusively on EP1 cyclones (most energetically active)
- All diagnostics are computed during the **intensification phase** when energy conversions are most active
- Multi-domain comparison ($5^\circ \times 5^\circ$, $15^\circ \times 15^\circ$, $30^\circ \times 30^\circ$) allows scale-dependent analysis
- Results will be compared with EP2 and EP3 in future work to understand what makes EP1 unique

Step 4: Scientific Notes and Implementation Details

Physical Constants and Quality Controls

Constants (SI units)

```
G = 9.80665           # Standard gravity (m s-2)
OMEGA = 7.292e-5       # Earth angular velocity (rad s-1)
R_d = 287.0            # Gas constant dry air (J kg-1 K-1)
C_p = 1004.0           # Specific heat constant pressure (J kg-1 K-1)
KAPPA = R_d / C_p      0.286 # Poisson constant
P_0 = 100000.0         # Reference pressure 1000 hPa (Pa)
R_EARTH = 6.371e6      # Earth radius (m)
```

Quality Control Thresholds

```
MIN_LAT = 5.0          # Minimum |lat| for EGR (avoid f=0 at equator)
MAX_EGR_DAY = 5.0       # Maximum reasonable EGR (cap unphysical values)
MIN_N_SQUARED = 1e-6    # Minimum N2 for stable stratification (s-2)
```

Eady Growth Rate (EGR) - Detailed Derivation

1. Virtual Temperature Correction

Standard temperature T does not account for moisture effects on air density. Virtual temperature T_v is the temperature dry air would need to have the same density as moist air at the same pressure:

$$T_v = T(1 + 0.61q)$$

where: - T = temperature (K) - q = specific humidity (kg/kg) - 0.61 comes from the ratio of gas constants: $(R_v/R_d - 1) \approx 0.608$

Physical meaning: Moisture makes air less dense (water vapor lighter than dry air), so virtual temperature is higher than actual temperature in moist air.

Typical magnitude: For subtropical cyclones with $q \sim 0.01$ kg/kg, correction is ~ 6 K.

2. Virtual Potential Temperature

Potential temperature θ is the temperature an air parcel would have if brought adiabatically to reference pressure $p_0 = 1000$ hPa:

$$\theta = T \left(\frac{p_0}{p} \right)^\kappa$$

Virtual potential temperature combines both corrections:

$$\theta_v = T_v \left(\frac{p_0}{p} \right)^\kappa$$

Physical meaning: θ_v is conserved for adiabatic, reversible processes in moist air. It's the natural vertical coordinate for atmospheric stability.

3. Brunt-Väisälä Frequency (Static Stability)

The Brunt-Väisälä frequency N characterizes the frequency of vertical oscillations of a displaced air parcel in a stably stratified atmosphere:

$$N^2 = \frac{g}{\theta_v} \frac{\partial \theta_v}{\partial z}$$

Derivation (simplified): 1. Vertically displace air parcel by z in environment with stratification $\partial\theta_v/\partial z$ 2. Parcel conserves its θ_v , environment has different θ_v 3. Buoyancy force: $F = -g\frac{\delta\theta_v}{\theta_v}$ 4. Equation of motion: $\frac{d^2(\delta z)}{dt^2} = -N^2(\delta z)$ 5. Solution: oscillation with frequency N (if $N^2 > 0$)

Physical interpretation: - $N^2 > 0$: Stable stratification $\rightarrow$ parcel oscillates (gravity waves) - $N^2 = 0$: Neutral stratification $\rightarrow$ parcel remains displaced - $N^2 < 0$: Unstable stratification $\rightarrow$ parcel accelerates away (convection)

Typical values: - Troposphere: $N \sim 0.01 \text{ s}^{-1}$ (period $\sim 10 \text{ min}$) - Stratosphere: $N \sim 0.02 \text{ s}^{-1}$ (period $\sim 5 \text{ min}$) - Our results: $N \sim 0.0105 \pm 0.0020 \text{ s}^{-1}$

Computational implementation:

```
# 3-level centered finite difference
dtheta_v = theta_v[upper] - theta_v[lower]
dz = z[upper] - z[lower]
dtheta_v_dz = dtheta_v / dz

# N^2 = (g/v)(v/z)
N_squared = (G / theta_v[middle]) * dtheta_v_dz
N = sqrt(N_squared) # only if N^2 > MIN_N_SQUARED
```

4. Vertical Wind Shear

Magnitude of the vertical wind shear vector:

$$\left| \frac{\partial \vec{V}}{\partial z} \right| = \sqrt{\left(\frac{\partial u}{\partial z} \right)^2 + \left(\frac{\partial v}{\partial z} \right)^2}$$

Physical meaning: Rate of change of horizontal wind with height. Strong shear tilts developing baroclinic waves, allowing efficient extraction of potential energy.

Thermal wind balance: In geostrophic balance, vertical shear is proportional to horizontal temperature gradient:

$$\frac{\partial \vec{V}_g}{\partial z} = -\frac{g}{fT} \hat{k} \times \nabla_p T$$

This links baroclinicity (temperature gradient) to vertical shear $\rightarrow$ both appear in EGR formula.

Typical values: - Weak shear: $< 0.003 \text{ s}^{-1}$ - Moderate: $0.003\text{-}0.006 \text{ s}^{-1}$ - Strong: $> 0.006 \text{ s}^{-1}$ - Our results: $0.0051 \pm 0.0015 \text{ s}^{-1}$ (moderate to strong)

5. Eady Growth Rate Formula

$$\sigma_{EGR} = 0.31 \frac{|f|}{N} \left| \frac{\partial \vec{V}}{\partial z} \right|$$

Derivation: From Eady (1949) model of baroclinic instability on an f-plane with constant N and linear shear. The factor 0.31 comes from solving the eigenvalue problem for the most unstable wave.

Dimensional analysis: - $[f] = s^{-1}$ (frequency) - $[N] = s^{-1}$ (frequency) - $[\partial V / \partial z] = s^{-1}$ (shear) - $[\sigma] = \frac{s^{-1}}{s^{-1}} \cdot s^{-1} = s^{-1}$ (growth rate)

Physical interpretation:

1. $|f|$ (**rotation**): Provides restoring force for geostrophic balance. Higher latitude $\rightarrow$ stronger rotation $\rightarrow$ faster growth.
2. $1/N$ (**weak stability**): Weaker static stability allows easier vertical displacement $\rightarrow$ faster growth. Factor $1/N$ means growth rate inversely proportional to stability.
3. $|\partial V / \partial z|$ (**shear**): Tilts the wave, allowing efficient conversion of available potential energy to kinetic energy.

Why 0.31? From Eady model: - Most unstable wavelength: $\lambda = 3.9 \frac{NH}{f}$ (H = scale height) - Maximum growth rate: $\sigma_{max} = 0.31 \frac{f}{N} |\partial V / \partial z|$

Conversion to day⁻¹:

$$\sigma_{day} = \sigma_{s^{-1}} \times 86400 \text{ s/day}$$

Interpretation of values: - 0.5-1.0 day⁻¹: Moderate baroclinic instability - 1.0-2.0 day⁻¹: Strong instability (typical cyclogenesis) - 2.0-5.0 day⁻¹: Very strong instability (explosive cyclogenesis) - > 5.0 day⁻¹: Likely unphysical (numerical issues or extreme events)

Our results: - Mean: 1.17-1.35 day⁻¹ (strong baroclinic instability) - Max: 2.44-4.11 day⁻¹ (very strong in most intense regions)

Rayleigh-Kuo (RK) Criterion - Detailed Derivation

1. Relative Vorticity

Vertical component of vorticity (rotation rate about vertical axis):

$$\zeta = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$$

Spherical coordinates (Earth is not flat!):

$$\frac{\partial}{\partial x} = \frac{1}{R_E \cos \phi} \frac{\partial}{\partial \lambda}$$

$$\frac{\partial}{\partial y} = \frac{1}{R_E} \frac{\partial}{\partial \phi}$$

where ϕ = latitude, λ = longitude.

Physical meaning: Measures local spinning of air. Positive for counterclockwise rotation (cyclonic in NH, anticyclonic in SH).

Typical values: - Synoptic systems: $|\zeta| \sim 10^{-5} \text{ s}^{-1}$ - Strong cyclones: $|\zeta| \sim 10^{-4} \text{ s}^{-1}$

2. Absolute Vorticity

Sum of relative vorticity and planetary vorticity:

$$\eta = \zeta + f$$

where $f = 2\Omega \sin \phi$ is the Coriolis parameter.

Physical meaning: Total rotation rate as seen from inertial (non-rotating) frame. Combines Earth's rotation (f) with atmospheric rotation (ζ).

Why “absolute”? It's the vorticity in an absolute (inertial) reference frame, not rotating with Earth.

At 45°S: $f \approx -10^{-4} \text{ s}^{-1}$ (negative in SH)

3. Rayleigh-Kuo Criterion

Statement: A necessary condition for barotropic instability is that the meridional gradient of absolute vorticity **changes sign** somewhere in the domain:

$$\frac{\partial \eta}{\partial y} = 0 \text{ somewhere, with } \frac{\partial \eta}{\partial y} < 0 \text{ on one side and } > 0 \text{ on other}$$

Or more simply: $\min(\partial \eta / \partial y) < 0$ AND $\max(\partial \eta / \partial y) > 0$

Historical context: - **Rayleigh (1880):** For inviscid parallel shear flow, instability requires $\frac{d^2 U}{dy^2}$ to change sign - **Kuo (1949):** Extended to rotating fluids: criterion becomes sign change of $\frac{\partial \eta}{\partial y}$

Physical mechanism:

1. **Basic state:** Flow with absolute vorticity gradient $\partial\eta/\partial y$
2. **Perturbation:** Small wave-like disturbance
3. **Vorticity equation:** Perturbation vorticity evolves as $\frac{\partial\zeta'}{\partial t} + V\frac{\partial\zeta'}{\partial y} = -v'\frac{\partial\eta}{\partial y}$
4. **Growth:** If $\partial\eta/\partial y$ changes sign, perturbations can extract energy from basic flow

Why sign change? Counter-propagating Rossby waves on the two sides of the $\partial\eta/\partial y = 0$ point can phase-lock and amplify each other $\rightarrow$ instability.

Interpretation: - Criterion is **necessary** but **not sufficient** - Satisfaction means instability is *possible* (if other conditions met) - Non-satisfaction means instability is *impossible*

4. Connection to Barotropic Energy Conversion

Barotropic conversion C_k represents kinetic energy transfer from eddies to mean flow (or vice versa):

$$C_k = -\overline{u'v'}\frac{\partial\bar{u}}{\partial y} - \overline{v'v'}\frac{\partial\bar{v}}{\partial y}$$

Physical link to RK: - RK criterion indicates regions where wave growth can occur via horizontal shear - Growing waves generate Reynolds stresses $\overline{u'v'}$ - These stresses cause momentum transport and energy conversion - Negative C_k (common in cyclones) means energy flows from mean to eddies

Our finding: All 10 EP1 cyclones satisfy RK criterion at all scales (5°, 15°, 30°).

Implication: Favorable conditions for barotropic processes are widespread, not just locally confined.

Results Interpretation

Eady Growth Rate Statistics

Domain	Size	Mean EGR (day ⁻¹)	Std Dev	Max EGR (day ⁻¹)
Local	5°	1.237	0.441	2.442 ± 0.668
Mesoscale	15°	1.353	0.295	3.518 ± 0.673
Synoptic	30°	1.169	0.178	4.111 ± 0.643

Key findings: 1. **Strong baroclinic instability:** EGR $\sim$ 1.2-1.4 day⁻¹ indicates rapid growth potential 2. **Scale consistency:** EGR values similar across scales $\rightarrow$ baroclinic processes are large-scale 3. **Spatial structure:** Maximum

EGR increases with domain size (local max < mesoscale max < synoptic max)
 - Suggests strongest instability not concentrated at cyclone center - Rather, extends across broader synoptic environment 4. **Variability:** Higher variance at local scale (=0.44) than synoptic (=0.18) - Local conditions more variable case-to-case - Large-scale baroclinicity more consistent

Comparison with literature: - Gyakum et al. (1989): EGR $\sim 1\text{-}2 \text{ day}^{-1}$ for explosive cyclogenesis - Our values (1.2-1.4 day^{-1}) consistent with intensifying extratropical cyclones

Rayleigh-Kuo Criterion Results

Domain	Size	Cases Satisfied	Percentage
Local	5°	10/10	100%
Mesoscale	15°	10/10	100%
Synoptic	30°	10/10	100%

Key findings: 1. **Universal satisfaction:** All EP1 cyclones meet necessary condition for barotropic instability 2. **Scale independence:** Criterion satisfied at all scales examined 3. **Contrary to hypothesis:** Initial expectation was local-scale phenomenon - Results show large-scale flow configuration supports barotropic processes - Suggests interaction between synoptic forcing and local cyclone dynamics

Physical interpretation: - Large-scale flow pattern (jets, fronts) creates $\partial\eta/\partial y$ structure - This structure persists across scales from cyclone core to synoptic environment - Both local (cyclone-induced) and remote (synoptic) contributions to vorticity gradient

Connection to Ck: - Mean Ck for EP1: -16.48 W/m^2 (strongest barotropic conversion of all EPs) - RK satisfaction indicates this conversion occurs through barotropic instability mechanism - Negative Ck means energy flows from mean flow $\rightarrow$ eddies $\rightarrow$ consistent with RK-based wave growth

Computational Considerations

Vertical Derivatives

Centered finite differences used for accuracy:

$$\frac{\partial f}{\partial z} \approx \frac{f_{i+1} - f_{i-1}}{z_{i+1} - z_{i-1}}$$

Requires 3 vertical levels (upper, middle, lower).

Advantages: - Second-order accurate: $O(\Delta z^2)$ - Symmetric: no bias toward upper or lower level

Pressure to height conversion:

$$z = \frac{\Phi}{g}$$

where Φ = geopotential ($\text{m}^2 \text{s}^{-2}$), downloaded from ERA5.

Horizontal Derivatives

Spherical geometry accounted for using metric factors:

```
dx = R_EARTH * cos(lat) * dlon_radians  
dy = R_EARTH * dlat_radians
```

```
du_dy = gradient(u, axis=0) / dy  
dv_dx = gradient(v, axis=1) / dx
```

Why necessary? - On sphere, 1° longitude 1° latitude in metric distance -
At equator: 1° lon 111 km, 1° lat 111 km - At 45° : 1° lon 78 km, 1° lat 111 km

Failure to account for this introduces systematic errors in vorticity calculation.

Quality Control Flags

1. **Near-equator masking:** $|lat| < 5^\circ$ excluded
 - $f \rightarrow 0$ near equator makes f/N term unstable
 2. **Static stability check:** Require $N^2 > 10^{-6} \text{ s}^{-2}$
 - Filters unstable stratification (convection, not baroclinic instability)
 - Also filters numerical noise in $\partial\theta_v/\partial z$
 3. **EGR cap:** Values $> 5.0 \text{ day}^{-1}$ set to NaN
 - Removes unphysical extremes from averaging
 - Can occur near topography, data gaps, or numerical issues
-

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